

MUHAMMAD LUTFI WIBOWO

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Fresh graduate Mechanical Engineering with hands on internship experience in Mechanical, Electrical, and Plumbing (MEP) engineering across commercial and industrial projects. Experienced in preparing Detailed Engineering Design (DED) drawings, plumbing, and fire protection systems using AutoCAD and GstarCAD while coordinating with multidisciplinary teams. Strong analytical, communication, and problem-solving skills with a strong interest in building services and MEP design engineering.

Work Experiences

PT. LEGUAN DESAIN INDONESIA (Intern) – Jakarta, Indonesia

Jan 2026 – Mar 2026

Mechanical CAD Drafter (MEP)

- Prepared and revised 20+ Detailed Engineering Design (DED) drawings for plumbing and fire protection systems using AutoCAD and GstarCAD, ensuring compliance with engineering standards and project specifications.
- Prepared ducting and piping layouts based on engineering calculations, technical specifications, and project requirements to support efficient MEP system design.

PT. PAKUWON SENTOSA ABADI Tbk (Intern) – Jakarta, Indonesia

Jun 2025 – Aug 2025

Mechanical, Electrical, and Plumbing Engineer

- Involved in HVAC installation works (AHU, ducting, diffusers), ensuring alignment with technical drawings, specifications, and site standards, while coordinating with field teams during installation phases.
- Conducted inspection and testing support for fire protection systems (sprinkler, hydrant, fire pump) to verify system readiness, performance, and compliance with applicable safety standards and regulations.
- Coordinated genset installation and ATS/AMF setup for backup power systems (up to 5000 kVA), ensuring proper system integration, functionality, and reliability during operational testing.
- Handled monitoring and verification of electrical installation, including MVMDP, LVMDP, and cabling systems (20 kV–400 V), ensuring installation accuracy, safety procedures, and compliance with project specifications.

PT. DHARMA SATYA NUSANTARA Tbk (Intern) – Temanggung, Indonesia

Jan 2025 – Feb 2025

Engineering and Development

- Monitored thermal oil boiler operation (3 ton/hour, 1800 kW) to maintain stable system performance and efficiency ($\geq 82\%$).
- Supported maintenance and troubleshooting of boiler components, piping systems (± 58 m), and control instruments.
- Conducted heat loss analysis on piping system, identifying energy losses of 3.75 kW.
- Collected and processed operational data (temperature up to 200°C , flow rate $178 \text{ m}^3/\text{h}$, pipe dimensions) for performance evaluation.

Education Level

Brawijaya University – Malang, Indonesia

2022 - 2026

- Bachelor of Mechanical Engineering – GPA: 3.33 / 4.00
- Undergraduate Thesis in The Potential of Ocean Current Energy in Indonesian Waters Based on an Analysis of Quarterly Satellite Data for the Period 2009–2024 Using the K-Means Clustering Method
- Relevant Courses: Thermodynamics, Fluid Mechanics, Heat Transfer, Energy Conversion, Combustion Technology, Renewable Energy.

Organisational and Committee Experiences

Himpunan Mahasiswa Mesin FT-UB, Malang

May 2025 – Apr 2026

Internal Staff (Sports Operations)

- Directed the end-to-end execution of 3+ major student sports programs, achieving a 50%+ participation rate and fostering high team engagement across the department.
- Coordinated with internal stakeholders and facility management to optimize resource allocation, ensuring all events met quality and safety standards.

PROBINMABA MESIN FT-UB, Malang

Aug 2024 – Des 2024

Field Operations & Crowd Control Lead

- Managed and directed over 270 personnel to form a secure perimeter during student mobilization, ensuring the privacy and order of the event from external interference.
- Orchestrated the movement of 200+ new students across campus routes, coordinating timing and path sterilization with security and other departments to prevent bottlenecks.
- Conducted security screenings and inspections to ensure a safe, secure, and orderly event environment.
- Enforced a strict code of conduct and standard operating procedures for the field team to ensure disciplined communication and professional conduct during operations.

Technical Skills

Mechanical & Utility Systems

- Industrial Boiler System Monitoring & Thermal Efficiency Analysis
- HVAC System Installation & Basic Maintenance
- Plumbing and Building Utility Systems
- Generator Set (Genset) Installation & Commissioning

Engineering Software

- AutoCAD & GstarCAD (MEP Drafting, DED Drawings & Technical Documentation)
- MATLAB (Data Analysis, Numerical Computation & Engineering Modeling)
- Autodesk Inventor (Mechanical Design & 2D Drawings)

Electrical Building Systems

- MVMDP and LVMDP Basic Installation Knowledge
- Building Electrical Distribution System

Soft Skills

Team Leadership | Cross-Functional Coordination | Communication Skills | Problem Solving | Attention to Detail | Stakeholder Management.

Languages

Indonesian (Native) | English (Intermediate)

Achievements

- YBM Brilliant Scholarship Awardee (Malang)
- Pancakarsa Scholarship Awardee (Kab. Bogor)
- Hafidz Al-Qur'an (30 Juz)